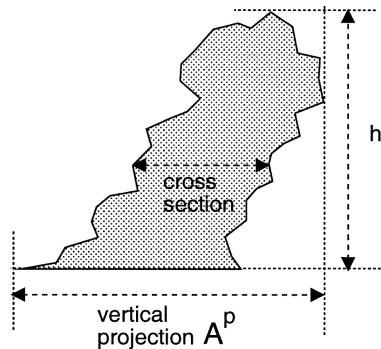


Large-eddy simulations (LES) of shallow cumulus: simulated clouds, not measurements. Each cloud gets one size, and the paper studies, among other things, the histogram of these sizes.



$$l = \sqrt{A^p}$$

$$N = \int_0^\infty \mathcal{N}(l) dl$$

Three simulated cases, set up like real campaigns:

- BOMEX: Barbados, ocean.
- SCMS: Florida, land.
- ARM: Oklahoma, land, diurnal cycle.

Symbols

cloud: group of touching saturated grid cells in 3D

A^p vertically projected area of one cloud: its footprint seen from above

l cloud size, m

$\mathcal{N}(l)$ number of clouds per unit size, m^{-1}

N total number of clouds in the domain

Find each cloud in 3D → project it vertically → histogram of the sizes l

Neggers et al. (2003): cloud-size statistics

Size Statistics of Cumulus Cloud Populations in Large-Eddy Simulations, Neggers, Jonker and Siebesma (2003). The study measures the sizes of shallow cumulus clouds in LES and their contributions to cloud cover and vertical transport. BOMEX represents steady trade-wind cumulus over the ocean. SCMS and ARM represent continental shallow cumulus, with a diurnal cycle in ARM. All clouds are simulated: each case is initialised and forced with measurements from a real campaign (Barbados, Cocoa Beach in Florida on 5 August 1995, Oklahoma SGP on 21 June 1997), but the simulated clouds are not the observed ones. Comparison with nature is through published exponents and dominant sizes. The reference horizontal grid spacing is 50 m, in a 6.4 km domain. Repeated snapshots sample tens of thousands of clouds per case.

A cloud is a connected set of saturated grid cells. Figure 3 shows the distinction between a cross-section at one height and the vertical projection of the whole cloud. The chosen size is $l = \sqrt{A^p}$, the side of a square with the cloud's projected area, not a radius or a diameter. It is chosen so that LES can be compared with satellite images, which see only projections. The size density $\mathcal{N}(l)$ gives counts per unit size, and its integral is the total count. The bottom line summarizes the actual measurement procedure. Source: Sections 2–3 and Appendix, Figure 3, Equations (1)–(2).

The histogram counts clouds per size: **small clouds are by far the most numerous**. Below a size ℓ_d their number falls as a power law of size; above ℓ_d it falls faster.

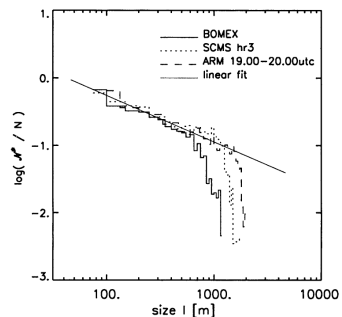


Fig. 4a: size density against l .

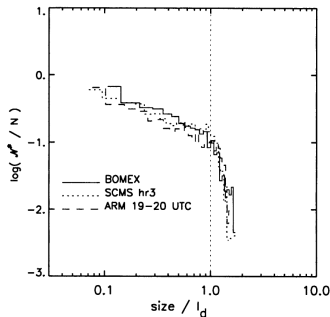


Fig. 6: against l/ℓ_d , collapsed.

$$\mathcal{N}(l) = al^b$$

$$b = -1.70 \text{ for } l < \ell_d$$

Symbols

ℓ_d scale break: size where the exponent changes. BOMEX 700 m, SCMS 1050 m, ARM 400 $\rightarrow$ 1250 m in the afternoon

a, b fit constants; b is the exponent
 $\mathcal{N}^* = l \ln(10) \mathcal{N}$, clouds per decade of size (plotted), slope $b + 1 = -0.70$

Same exponent in all cases: only the break ℓ_d changes

A power law below a variable size break

Figure 4a shows the normalized size density in logarithmic bins. Below the scale break, the three cases have approximately the same slope. The fitted log-bin slope is -0.70 , corresponding to the exponent $b = -1.70$ of $\mathcal{N}(l) = al^b$ because $\mathcal{N}^* = l \ln(10) \mathcal{N}$: a power law per decade has its exponent raised by one. The supported range spans roughly one decade between the grid scale and the break.

Above ℓ_d , the density decreases faster. Table 2 gives 700 m for BOMEX and 1050 m for SCMS. During the ARM afternoon, the break increases from 400 to 1250 m. Increasing the number of sampled clouds does not remove the break.

Figure 6 plots the three size densities against l/ℓ_d . Their approximate collapse supports a common functional form for these simulated cumulus populations, with a case-dependent break: the break is the only variable. Its physical origin remains unresolved; the paper mentions subcloud-layer depth and humidity fluctuations as candidates. It is a cloud-size scale, not an inter-cloud distance. Source: Section 4a, Figures 4a and 6, Table 2, Equations (8)–(9).

Weighting every cloud by its area gives the cloud cover contributed by each size. It grows up to ℓ_d and drops beyond, so **mid-size clouds cover most of the sky**.

$$\alpha^p(l) = \frac{l^2 \mathcal{N}(l)}{L_x L_y} \propto l^{b+2} = l^{0.30} \quad (l < \ell_d)$$

$$a^p = \int_0^\infty \alpha^p(l) dl$$

Symbols

$\alpha^p(l)$ cloud cover from clouds of size l , per unit size, m^{-1}

a^p projected cloud cover: fraction of the domain covered, seen from above

$L_x L_y$ horizontal area of the domain

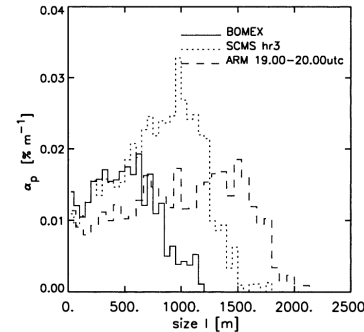


Fig. 7a: $\alpha^p(l)$, three cases.

The size break sets the dominant cloud cover

Figure 7a gives the contribution to projected cloud cover per unit size. Each cloud contributes its projected area l^2 , so $\alpha^p(l) = l^2 \mathcal{N}(l) / (L_x L_y)$, where $L_x L_y$ is the horizontal domain area. For a power law with exponent b , the cover decomposition scales as l^{b+2} . Below the break, $b = -1.70$ gives a positive exponent, 0.30: the many small clouds are too small to dominate. Above the break the density declines faster than l^{-2} , so the cover decomposition decreases. The peak therefore lies at an intermediate size near the break. Without a break, an exponent above -2 would let the largest clouds dominate.

This is a decomposition per unit linear size, not per logarithmic bin. The total projected cover is $a^p = \int_0^\infty \alpha^p(l) dl$. The paper notes that real shallow cumulus fields also show this intermediate dominant size. Source: Section 3, Section 4b, Figure 7a, Equations (3)–(4) and (10).

Cloud cover peaks near the break ℓ_d

Larger clouds rise faster, so mass flux still peaks at **intermediate** sizes, near the cover peak, but the smallest clouds drop out: their vertical velocity is very low.

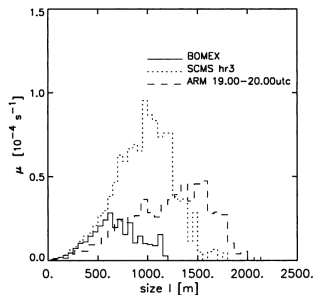
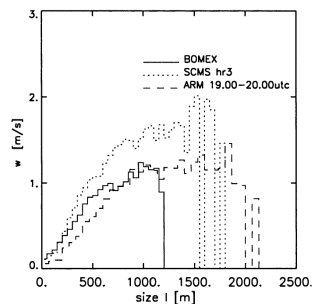


Fig. 7b: mean vertical velocity w .

Fig. 7c: mass flux $\mu(l)$.

Mass flux:

$$\mu(l, z) = \alpha(l, z) w(l, z)$$

$$\mu(l) = \frac{1}{h_c} \int_{h_c} \mu(l, z) dz$$

Symbols

$w(l, z)$ mean vertical velocity of clouds of size l at height z , m s^{-1}

$\alpha(l, z)$ their cloud cover at height z , per unit size

h_c depth of the cloud layer

μ mass flux divided by air density, per unit size, s^{-1}

Same intermediate dominant size as cover; the smallest clouds carry almost nothing

Mass flux also peaks at intermediate sizes

Figure 7b shows that cloud-mean vertical velocity generally increases with cloud size. Figure 7c shows the resulting mass-flux decomposition. At height z , $\mu(l, z) = \alpha(l, z)w(l, z)$, where $\alpha(l, z)$ is the cloud-fraction contribution per unit size at that height. Averaging over the cloud-layer depth h_c gives $\mu(l)$. Its integral over size is the total kinematic mass flux, in m/s under this paper's convention: mass flux divided by air density.

Cloud size is always the projected size, even when evaluating the flux at a specific height, so each cloud stays in one size bin at all heights. Since w increases with size, the small-cloud side is suppressed relative to the cover: the paper describes the dominant size as better defined and shifted somewhat toward larger sizes. Comparing Figs. 7a and 7c, the peak positions are nearly the same (read off the plots, about 650, 1000 and 1500 m for BOMEX, SCMS and ARM), so the shift is in the weight of the distribution, not in the position of its maximum. The smallest clouds contribute very little to total vertical transport, mainly because of their very low vertical velocities.

The intermediate dominant size persists for tested grid spacings of 25–100 m. A 3.2 km domain truncates the largest clouds. Increased shear broadens projected sizes while leaving total mass flux almost unchanged. These sensitivity tests support the main transport result but do not identify the mechanism of the size break. Source: Section 3, Section 4b–e, Figure 7b–c and Figures 9–11.

The paper gives no thermal statistics. Fig. 7b gives the mean updraft w of each cloud size: keep the sizes where w beats a pilot’s sink rate. Only the two land cases, SCMS and ARM, are used.

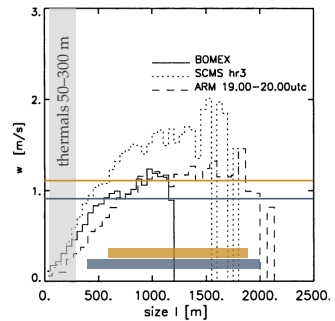


Fig. 7b; overlay ours.

“Climbable clouds”: 0.4–2 km. Expected thermals: 50–300 m

§4b, Fig. 7b, w read off the plot. Sink rates: typical straight-flight minimum sink. Thermal diameters: rough estimate, not from a measurement.

Climbing needs w above the sink rate

w : mean updraft of clouds of size l . Sink rates are for straight flight: a circling wing sinks faster, so **the real min. sink thresholds are higher**.

	min. sink	clouds with $w \geq$ sink	thermal diameter
Paraglider	1.1 m/s	0.6–1.9 km	50–300 m
Hang glider	0.9 m/s	0.4–2.0 km	80–300 m

Clouds of 50–300 m are the most numerous, but their mean updraft stays below 0.7 m/s: too weak to keep a paraglider or a hang glider climbing.

Clouds with enough updraft are 0.4–2 km wide: several times, up to ten times, the 50–300 m expected for the thermals pilots use.

“Climbable clouds” are far wider than thermals

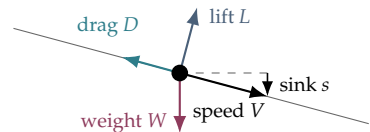
Neggers et al. give no statistics of thermals. Figure 7b gives, for each cloud size l , the cloud-mean vertical velocity w , which grows with size. A pilot climbs only where the air rises faster than the wing sinks, so the minimum sink rate sets a threshold on w : about 1.1 m/s for a paraglider and 0.9 m/s for a hang glider, in straight flight. Circling increases the sink rate, so these thresholds are lower bounds and the true climbable ranges would shift to even larger clouds.

Only the two land cases are used. Over the ocean (BOMEX) the surface sensible heat flux is 8 W m^{-2} , against 150 W m^{-2} in SCMS and up to 140 W m^{-2} in ARM (Table 1): without a heated ground, the dry thermals pilots use are weak.

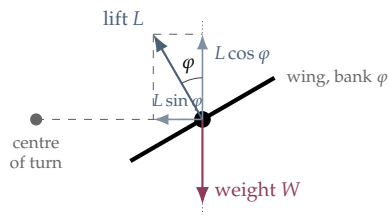
Read off the plot, SCMS clouds exceed 0.9 m/s from about 400 m and 1.1 m/s from about 600 m, up to the largest clouds near 1.8 km. ARM clouds exceed 0.9 m/s from about 700 m and 1.1 m/s from about 850 m, up to about 2 km. Taking both cases, the climbable cloud sizes are 0.6–1.9 km for a paraglider and 0.4–2.0 km for a hang glider. The expected thermal diameters are 50–300 m and 80–300 m. Clouds of that size have w well below either sink rate.

The ranges start above the widest expected thermal and reach 2 km. If cloud sizes were read as thermal sizes, the thermals would come out several times too wide: l is the square root of the projected area, which merges all heights (Fig. 3), and a cloud is fed through its base by a narrower thermal. Caveats: w is the in-cloud mean, which includes weak edges and differs from the updraft below cloud base; the statistics are LES snapshots of two land cases. Source: Section 4b, Figure 7b, Table 1.

Why circling sinks faster than straight flight, at the same angle of attack.



side view, straight glide



back view, turn

1 Energy. Lift does no work; drag uses DV per second, paid for by lost height

2 Glide ratio $E = L/D$ is set by the angle of attack, so $D = L/E$

3 Straight flight: lift $\approx$ weight, as the path is only $\approx 6^\circ$ below horizontal

4 Turn: the vertical part of lift holds the weight, $L \cos \varphi = W$

5 Same angle of attack: lift grows as V^2

$$Ws = DV$$

$$s = VL/(EW)$$

$$L \approx W, \quad s_0 \approx V_0/E$$

$$L = W/\cos \varphi$$

$$V = V_0/\sqrt{\cos \varphi}$$

$$\text{Put 3, 4, 5 into 2: } s = \frac{V_0}{\sqrt{\cos \varphi}} \cdot \frac{W}{\cos \varphi} \cdot \frac{1}{EW} = \frac{s_0}{(\cos \varphi)^{3/2}}$$

Circling sinks faster: $\times 1.10$ at 20° of bank, $\times 1.24$ at 30° , $\times 1.68$ at 45°

Steady flight; small glide angle, so $L \approx W$ (error below 1% for a glide ratio of 9). The extra drag of turning is neglected, so the true factor is larger.

Appendix: sink rate in a banked turn

Derivation of the sink rate in a steady banked turn, used on the previous slide. Step 1: in a steady glide the speed is constant and the lift, perpendicular to the velocity, does no work; the drag dissipates DV per second, and the only source paying for it is height, so $Ws = DV$. Step 2: the glide ratio $E = L/D$ depends only on the angle of attack, so $D = L/E$ and $s = VL/(EW)$. Step 3: in straight flight the lift equals the weight, $L = W$, so $s_0 = V_0/E$.

Step 4: in a turn banked by φ the lift is tilted towards the centre of the turn. Its vertical part must still hold the weight, $L \cos \varphi = W$, so $L = W/\cos \varphi$; its horizontal part, $L \sin \varphi$, bends the path. Step 5: at the same angle of attack the lift grows as the square of the speed, $L = \frac{1}{2}\rho V^2 SC_L$, so a lift $1/\cos \varphi$ times larger needs $V = V_0/\sqrt{\cos \varphi}$, and E is unchanged. Putting 3, 4 and 5 into 2 gives $s = s_0/(\cos \varphi)^{3/2}$: 10% more at 20° of bank, 24% at 30° , 68% at 45° .

The factor does not depend on the angle of attack: at every angle of attack the turn multiplies the straight-flight sink by $1/(\cos \varphi)^{3/2}$, so the minimum sink in a turn is the straight-flight minimum sink times the same factor. Steps 3 and 4 use $L \approx W$, valid for a small glide angle: with a glide ratio of 9 the path is about 6° below horizontal and the error is below 1%. Real wings, paragliders in particular, also lose more to turning drag. The straight-flight minimum sink is therefore a lower bound on the updraft needed to climb while circling. The two force balances also give the turn radius, $r = V^2/(g \tan \varphi)$: circling in a narrow thermal needs a steep bank, hence a higher sink rate.